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SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*
Assessment Tool Weightage: 40%

Annexure A
SHORT ANSWER TEST

Description about the assessment tool:
Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Code of Conduct:

I T. Subahan (192511121) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	2	2	2	2
Organization	15%	1	1	1	
Accuracy	10%				
Clarity	5%				

Signature of the Faculty

Total Marks 27 **/40**

2. Sliding Window Protocol:

Sliding window size = 6 frames

frame size = 1024 bytes

Total frames = 48

No of transmission windows = $\frac{48}{6}$ = 8 windows

Total retransmission = 8 x 1 = 8 frames

Transmission window required = 8

Total retransmissions = 8 frames

Flow control ensures that the sender does not send data faster than the receiver can process. this prevent buffer overflow, reduces packet loss and improves communication efficiency.

3. IP Packet Fragemantation:

IP Packet size = 12,000 bytes

MTU = 1500 bytes

IP header per fragment = 20 bytes

Max data $= 1500 - 20 = 1480$ bytes
No of fragements $= \lceil 12000 \div 1480 \rceil = 9$ fragements
Total header overhead $= 9 \times 20 = 180$ bytes.

The Network layer divides large packets into fragements assigns fragement information (ID, offset, and flags) routes them independently and enables the destination to reassamble th into orginal packet correctly.

4. Sliding Window Protocol:

Sliding window size $= 6$ frames
Frame size $= 1024$ bytes
Total frames $= 48$

Number of transmission windows $= \frac{48}{6} = 8$ wi
one frame is lost in each window
Total retransmission $= 8$ frames

Flow control regulates the rate of d transmission b/w sender and reciver, pr congestion and improving reliable communi

Session Layer Synchronization:

Video file size = 600 MB

Checkpoint after every 60 MB

Failure after 390 MB

Checkpoints before failure = $\frac{390}{60}$ = 6 checkpoints

(at 60, 120, 180, 240, 300, 360 MB).

Last checkpoint = 360MB

Data to retransmit = 390 − 360 = 30MB

Checkpoints created before failure = 6

Data to restransmit after recovery = 30MB

Importance of Synchronization: Synchronization check-points allow communication to resume from the last saved checkpoint instead of restarting the entire transfer. this saves time, bandwith, and improves reliability.